DATA INTERPRETATION-CO3-AT3

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COURSE NAME: REINFORCEMENT LEARNING

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1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

Step 1 – Observe the trend:
The reward increases gradually from **10 in the early episodes to 45 in later episodes**.

Step 2 – Meaning of increasing reward:
A higher reward means the agent is making **better decisions** and receiving more positive outcomes from the environment.

Step 3 – Learning performance:
The gradual increase indicates that the agent is **learning from its previous experiences** and improving its actions.

Step 4 – Policy improvement:
The agent's policy is improving because it is increasingly selecting actions that lead to higher rewards.

Step 5 – Conclusion:
Therefore, the increasing reward from 10 to 45 indicates **successful learning and policy improvement**. If the reward eventually becomes stable around a high value, it can indicate that the agent is approaching a good or optimal policy.

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2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better

performance.

Step 1 – Understand epsilon:
Epsilon (ε) controls the balance between **exploration and exploitation**.

- **Exploration:** Trying new actions.
- **Exploitation:** Selecting the action currently believed to be the best.

Step 2 – ε = 0.1:
The agent explores only about 10% of the time and exploits most of the time. Therefore, it mainly selects actions that have already produced good rewards.

Step 3 – ε = 0.3:
The agent explores more frequently. This may help discover new actions but also causes some loss of reward by selecting less-known actions.

Step 4 – ε = 0.5:
The agent explores very frequently. Half of its decisions may involve exploration, which can reduce the immediate cumulative reward.

Step 5 – Conclusion:
Since **ε = 0.1 gives the highest cumulative reward**, it provides the best performance in this particular situation. The agent benefits from exploiting its learned knowledge rather than spending too much time exploring.

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3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

Step 1 – Compare rewards:
The average rewards are measured at **100, 200 and 300 episodes**.

Step 2 – Observe the result:
Q-Learning consistently obtains **slightly higher rewards** than SARSA.

Step 3 – Q-Learning:
Q-Learning is an **off-policy** algorithm. It learns the value of the best possible next action, even if the agent's current behavior uses exploration.

Step 4 – SARSA:
SARSA is an **on-policy** algorithm. It learns according to the actions that the agent actually follows, including exploratory actions.

Step 5 – Conclusion:
Since Q-Learning produces higher rewards consistently over the given episodes, it shows **better performance in this experiment**. The observed faster improvement suggests that **Q-Learning converges faster** than SARSA for this particular environment.

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4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance.

Step 1 – Understand discount factor:
The discount factor **γ (gamma)** determines how much the agent values future rewards compared with immediate rewards.

Step 2 – γ = 0.2:
The agent gives high importance to immediate rewards and gives very little importance to future rewards.

Step 3 – γ = 0.5:
The agent considers both immediate and future rewards with moderate importance.

Step 4 – γ = 0.9:
The agent gives strong importance to future rewards. It can therefore learn actions that may produce smaller immediate rewards but larger rewards in the long term.

Step 5 – Conclusion:
Since the average reward increases as γ increases, **γ = 0.9 is the most suitable value** among the given choices. It provides better long-term reward optimization and produces better learning performance in this experiment.

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5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.

Step 1 – Initial performance:
At **50 episodes**, the success rate is only **45%**. This indicates that the agent has not learned the task effectively.

Step 2 – Training:
As the number of episodes increases, the agent receives more experience and learns from its interactions with the environment.

Step 3 – Improved performance:
At **200 episodes**, the success rate reaches **92%**.

Step 4 – Interpretation:
The increase from 45% to 92% shows a significant improvement in the agent's decision-making ability.

Step 5 – Conclusion:
A 92% success rate strongly indicates that the agent has learned a **high-quality policy**. However, it cannot be concluded with absolute certainty that the policy is mathematically optimal unless the maximum possible

performance or convergence is verified. The trend strongly suggests that the agent is **approaching an optimal policy**.

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6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability.

Step 1 – Understand learning rate:
The learning rate controls **how much the agent changes its learned values after receiving new information**.

Step 2 – Learning rate = 0.01:
The agent updates its knowledge slowly. This can provide stable learning but may make convergence **too slow**.

Step 3 – Learning rate = 0.1:
The agent makes large updates. Learning may be faster, but excessive updates can cause **instability or fluctuations**.

Step 4 – Learning rate = 0.05:
This value provides a balance between update speed and stability. The agent learns sufficiently quickly without making excessively large changes.

Step 5 – Conclusion:
Since **0.05 provides higher stability**, it is the best choice among the three. It offers a good **trade-off between convergence speed and learning stability**.

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7. A Q-learning agent is tested with different exploration strategies: ε-greedy, softmax, and random exploration. The cumulative reward is highest with ε-greedy. Compare the strategies and defend why ε-greedy achieves better performance in this scenario.

Step 1 – ε-greedy:
The agent usually chooses the best-known action but occasionally selects a random action for exploration.

Step 2 – Softmax:
Softmax assigns probabilities to actions based on their estimated values. Better actions receive higher probabilities, while weaker actions still have a chance of being selected.

Step 3 – Random exploration:
Random exploration selects actions without strongly considering their learned values. Therefore, it may waste many decisions on poor actions.

Step 4 – Compare rewards:
In this scenario, **ε-greedy produces the highest cumulative reward** because it effectively combines exploration with exploitation.

Step 5 – Conclusion:
ε-greedy performs better because it spends most of its decisions exploiting the best-known actions while still exploring occasionally. Therefore, **ε-greedy is the best strategy in this experiment**.

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8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy.

Step 1 – Initial performance:
Initially, the agent takes approximately **50 steps** to reach the goal. This indicates inefficient navigation.

Step 2 – Learning:
Through repeated episodes, the agent learns which actions and paths lead closer to the goal.

Step 3 – Improved performance:
After **300 episodes**, the number of steps decreases to only **15**.

Step 4 – Interpretation:
Fewer steps mean the agent has learned a **more efficient path** and is making better decisions.

Step 5 – Conclusion:
The reduction from 50 to 15 steps strongly indicates **policy improvement and learning convergence**. If the number of steps remains close to 15 in later episodes, it provides stronger evidence that the agent has converged to an optimal or near-optimal policy.

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9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.

Step 1 – Understand discount factor:
The discount factor determines how much importance SARSA gives to **future rewards**.

Step 2 – $\gamma = 0.4$:
The agent focuses mainly on immediate rewards and gives less importance to future outcomes.

Step 3 – $\gamma = 0.7$:
The agent gives moderate importance to both immediate and future rewards.

Step 4 – γ = 0.95:
The agent strongly values future rewards. It can therefore learn strategies that may require several steps before receiving a large reward.

Step 5 – Conclusion:
Since **γ = 0.95 produces the highest long-term reward**, it is the most effective value in this experiment. It allows SARSA to consider long-term consequences and make decisions that maximize future returns.

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10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.

Step 1 – Early episodes:
The rewards fluctuate during the initial episodes because the agent has **limited knowledge of the environment** and is still exploring different actions.

Step 2 – Learning process:
Through repeated interaction, the agent learns which actions produce better outcomes.

Step 3 – After 500 episodes:
The reward becomes more stable. This indicates that the agent's decisions are becoming **more consistent**.

Step 4 – Policy improvement:
Stable rewards suggest that the agent has learned a policy that produces relatively predictable and good outcomes.

Step 5 – Conclusion:
The stabilization after 500 episodes indicates **policy improvement and learning stability**. In a stochastic environment, some variation may still remain, but a stable high-reward trend indicates that the learned policy is becoming **robust and approaching convergence**.